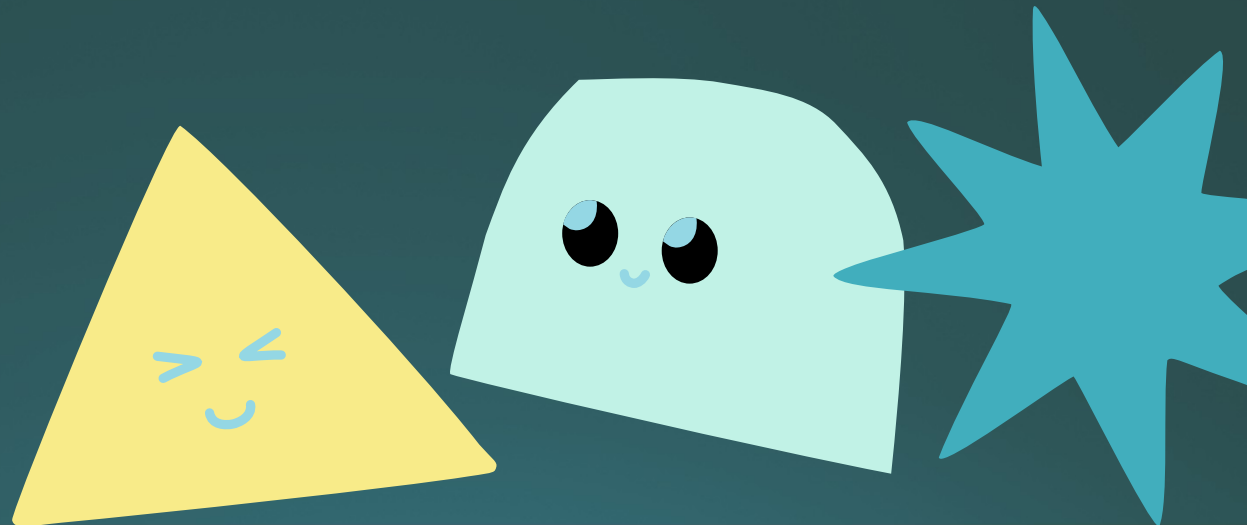


Engineering exploration for computer science and engineering



WHAT IS ENGINEERING?

- Engineering is the creative application of mathematics, science, and technology to design, build, and optimize solutions for real-world problems. It bridges the gap between scientific discovery and practical application, balancing functionality, cost, and safety to improve daily life



DIFFERENCE BETWEEN SCIENCE AND ENGINEERING.

science

Science is a discipline that studies and observes the natural world to understand how things work. It aims to build knowledge and make testable predictions about natural phenomena by formulating and testing hypotheses.

Engineering

Engineering is the application of scientific and mathematical principles to design, build, and maintain structures, machines, and systems. It involves solving practical problems by creating efficient and innovative solutions through the use of technology and materials.



DIFFERENCE BETWEEN SCIENTIST AND ENGINEERING.

Scientist

Scientists conduct studies and research to deepen their understanding of different topics, such as plants, animals, viruses, medicine, and planetary systems.

Example; A Physicist.

Engineer

Engineers invent, design, and build complex systems, technologies, and structures. As an engineer, you use mathematical principles and scientific findings to help you solve mostly real-world problems.

Example; A mechanical engineering.



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Explain the needs and wants of today's engineers (or modern engineers).

Needs:

- Strong technical knowledge.
- Problem-solving and analytical skills.
- Good communication and teamwork.
- Knowledge of modern technologies.
- Continuous learning and ethical responsibility.

Wants:

- Career growth and job security.
- Good salary and work-life balance.
- Opportunities for innovation and learning.
- Recognition and global exposure.



List and explain the various disciplines of engineering. Also, map the role of computer science engineering with other branches of engineering using real-world products as case studies.

Engineering Disciplines:

Computer Science Engineering (CSE) – Software and computing.

Mechanical Engineering – Machines and manufacturing.

Civil Engineering – Buildings and infrastructure.

Electrical Engineering – Power systems.

Electronics & Communication Engineering (ECE) – Electronics and communication.

Role of CSE with Other Branches:

Mechanical – Robotics and automation.

Civil – Smart city and traffic management software.

Electrical – Smart grids and power monitoring.

ECE – Smartphones and embedded systems.

Biomedical – Medical imaging and health monitoring



Write short notes on the misconceptions about Computer Science (CS) engineering.

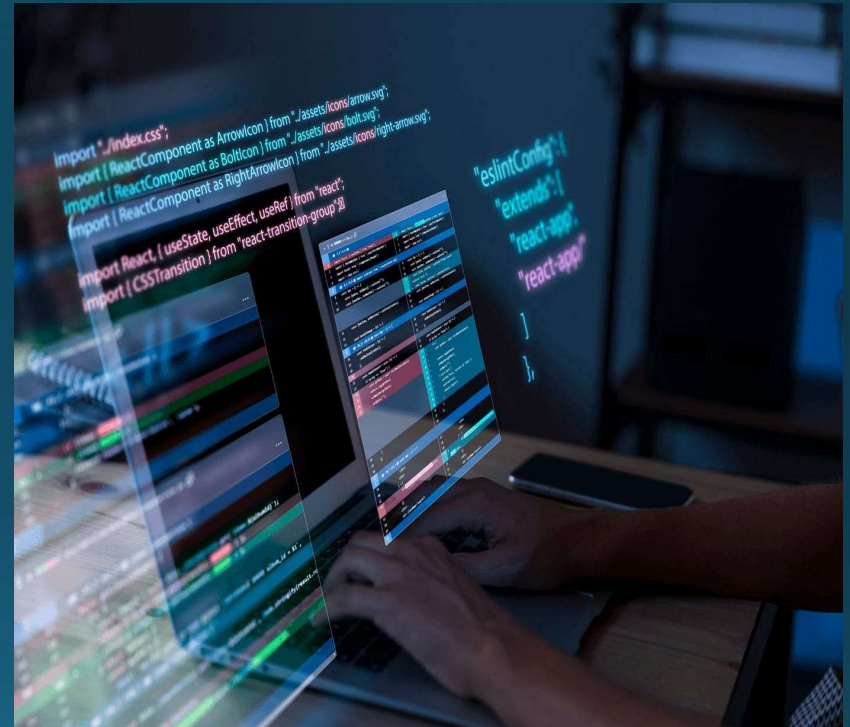
CS is not only about coding.

Good mathematics helps but is not the only requirement.

CS engineers have many career options besides software development.

Learning programming alone is not enough; core concepts are also important.

AI will create new opportunities, not replace all CS jobs.



What skills are required for a Computer Science (CS) engineering student?

Programming skills (C, C++, Java, Python).

Problem-solving and logical thinking.

Knowledge of data structures and databases.

Communication and teamwork.

Time management and creativity.

Continuous learning and adaptability.

Basic knowledge of AI, cloud computing, and cybersecurity.



THANK YOU

BY KARUPPUSAMY

